



GCN-BBB: Deep Learning Blood-Brain Barrier (BBB) Permeability Pharmacodynamics with Graph Convolutional Neural (GCN) Network

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Abstract

The Blood-Brain Barrier (BBB) is a selective barrier between the Central Nervous System (CNS) and the peripheral system, regulating the distribution of molecules. BBB permeability has been crucial in CNS-targeting drug development, such as glioblastoma-related drug discovery. In addition, more CNS diseases still present significant challenges, for instance, neurological disorders like Alzheimer's Disease (AD) and drug abuse. Conversely, cannabinoid drugs that do not cross the BBB are needed to avoid off-target CNS psychotropic effects. *In vitro* and *in vivo* experiments measuring BBB permeability are costly and low throughput. Computational pharmacodynamics modeling, particularly using deep-learning Graph Neural Networks (GNNs), offers a promising alternative. GNNs excel at capturing intricate relationships in graph-based information, such as small molecular structures. In this study, we developed GNNs model for BBB permeability using the graph representation of drugs. The GNNs were compared with other algorithms using molecular fingerprints or physical–chemical descriptors. With a dataset of 1924 molecules, the best GNNs model, a convolutional graph neural network using a normalized Laplacian matrix (GCN_2), achieved a precision of 0.94, recall of 0.96, F1 score of 0.95, and MCC score of 0.77. This outperformed other machine learning algorithms with molecular fingerprints. The findings indicate that the graphic representation of small molecules combined with GNNs architecture is powerful in predicting BBB permeability with high accuracy and recall. The developed GNNs model can be utilized in the initial screening stage for new drug development.

Keywords BBB · Deep learning · Graph Neural Network (GNN) · Permeability · Pharmacodynamics · Small molecule

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Introduction

The blood–brain barrier (BBB) is a physiological and biochemical shield separating the central nervous system (CNS) from peripheral tissues. It functions as the main active boundary between the fluctuating blood conditions of the CNS and the surrounding extracellular fluid (1) (2), preventing neurotoxic plasma components, blood cells, and pathogens from entering the brain (3). This selective permeability helps maintain CNS homeostasis (4, 5). The BBB comprises specialized microvasculature, including endothelial cells, basement membrane, pericytes, and astrocytes (6). Molecules cross the BBB through several pathways: paracellular diffusion, transcellular diffusion, and carrier/receptor-mediated transcytosis (7). Most molecules moving between the brain and the vascular system primarily utilize transcellular transport (Fig. 1). Some small molecules can enter the brain through passive diffusion, with hydrophilic molecules crossing at tight junctions between capillary endothelial cells (paracellular) and lipophilic molecules

following an osmotic gradient through the plasma membrane (transcellular) (8). In contrast, certain larger molecules, such as nanobodies or antibody drugs, gain access to the brain via receptor-mediated transcytosis (9, 10). Additionally, the BBB assists in the removal of metabolites and toxins from the CNS by utilizing active efflux transport proteins, including P-glycoprotein (P-gp, ABCB1), breast cancer resistance protein (BCRP, ABCG2), and multidrug resistance-associated proteins (MRP1-6, ABCC1-6), which are also known to play crucial roles in peripheral conditions affecting drug resistance and treatment efficacy, as observed for multiple myeloma and relapsed/refractory multiple myeloma (RRMM) (11, 12) and kidney fibrosis diseases (13). These transporters, which play critical roles in controlling the composition and volume of the cerebrospinal fluid, also actively transport drugs back into the blood by the use of ATP, and, thus, represent a significant factor in preventing drug entrance into the CNS (4, 14, 15).

Neurological Disorder Drug Development and the Role of the BBB in Drug Delivery

BBB penetration prediction has already played a significant role in glioblastoma-related drug discovery, in which small-molecule drugs are the primary therapeutic strategy. However, neurological disorders (NDs), such as stroke and Alzheimer's disease (AD), are major global health challenges affecting nearly 60 million individuals worldwide (16). By 2050, it is expected that over 139 million people will be living with dementia globally (17). In 2024, 6.9 million U.S. residents over 65 suffered from AD dementia, with this figure expected to double by 2060 (18). NDs rank among the top causes of death, accounting for 10.8% of deaths in the U.S. and 5.5% globally, with nearly 10 million deaths annually (19, 20). As life expectancy increases, the burden of these disorders will continue to grow (21).

The market for NDs treatments reached \$117.6 billion in 2022, with projections exceeding \$205.3 billion by 2033 (22). However, developing effective therapies remains difficult – particularly for AD, where most new drug candidates fail (23–27). A key challenge is the BBB (1), which blocks over 98% of small-molecule drugs from reaching the brain (28). Both water-soluble and lipophilic drugs face barriers to CNS entry, complicating the development of neurological treatments (8, 29). Despite the increasing focus on biologics, small molecules remain vital in Alzheimer's disease (AD) research and treatment. FDA-approved acetylcholinesterase inhibitors, such as Donepezil, Galantamine, and Rivastigmine, continue to be mainstays for managing AD symptoms. Additionally, promising small-molecule candidates, such as BACE1 inhibitors like LY2886721 and AZD3293, underscore the ongoing importance of small molecules in targeting disease-modifying pathways. Optimizing BBB

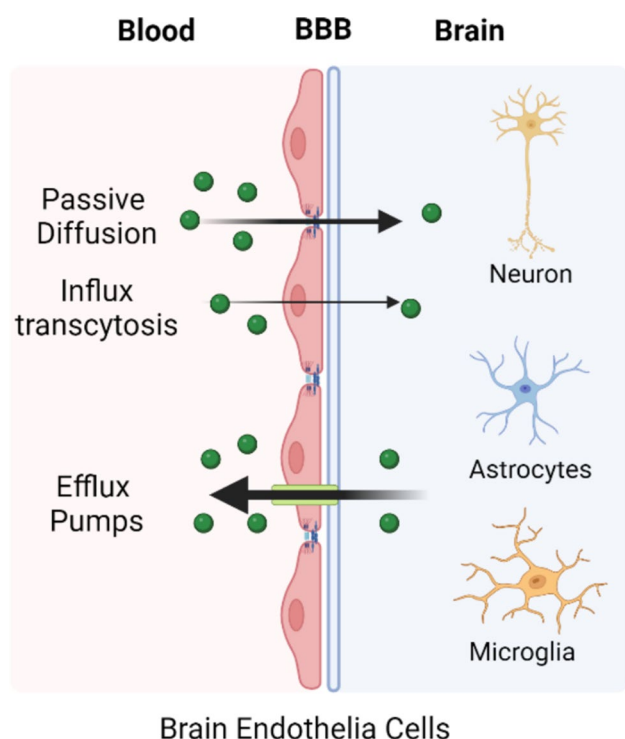


Fig. 1 A simplified illustration of transport routes for molecules/drugs to cross the BBB. This figure depicts the transport mechanisms that facilitate or impede the movement of molecules and drugs across the BBB. Small lipophilic molecules can cross the BBB at tight junctions between the endothelial cells (passive diffusion). Certain larger molecules and peptides can bind to specific receptors and be transferred through vesicles (influx transcytosis). Efflux pumps: P-glycoprotein can transport various drugs and endogenous compounds to reduce their penetration into the brain

permeability during drug discovery is therefore crucial for developing effective neurodegenerative disease therapies and enhancing CNS drug delivery.

Non-CNS-Targeting Drug Development – the Case of Cannabinoid Receptor 2

Drugs of abuse like cocaine, methamphetamine, and alcohol disrupt the BBB, increasing permeability and allowing toxins into the brain, causing neurotoxicity and addiction-related brain damage (30). Recent advancements in cannabinoid drug development focus on targeting peripheral cannabinoid receptors, particularly cannabinoid type 2 receptor (CB2/CNR2) expressed in immune cells (31), while intentionally designed to be BBB-excluded from the brain. This strategy affords potential therapeutic benefits for diseases such as kidney fibrosis (32), osteoporosis (33, 34), and inflammatory bowel disease (35), while avoiding deleterious psychotropic effects caused by low selectivity interactions with CNS-associated CB1 receptors (36). Thus, leveraging drug BBB permeability is a strategy that optimizes treatment outcomes for neurological and inflammatory conditions while minimizing CNS-related side effects (37).

In Vitro and In Vivo Methods for Assessing BBB Permeability

Scientific interest in the physiology and pathology of the BBB has driven the development of numerous experimental models. Accurately estimating the BBB permeability of CNS-targeting drug candidates is essential to accelerate neurological drug development. While various *in vitro* and *in vivo* methods exist for evaluating brain penetration (38–43), the increased experimental complexity raises costs and reduces throughput, making these approaches expensive and time-consuming. In contrast, computational pharmacanalytics offers a cost-effective and efficient alternative, complementing wet lab approaches by optimizing molecular design early in the drug discovery process. Leveraging these methods can improve success rates and significantly reduce the time and expense associated with more traditional ND/CNS drug development. Thus, adopting computational strategies to assess BBB permeability is crucial for enhancing the efficiency of therapeutic discovery.

In Silico Pharmacanalytics Methods to Predict BBB Permeability

Several computational approaches have been developed to predict BBB permeability (28, 29, 32, 33, 35, 36). Early methods, such as Pfizer's CNS multiparameter optimization (CNS MPO) score, used simple rules to identify CNS-targeting molecules (44). Traditional QSAR/QSPR models

also predict metrics like logBB, showing, for example, that CNS drugs generally have lower polar surface areas (PSA) than non-CNS drugs (45–47). However, BBB permeability often involves nonlinear relationships that traditional models struggle to capture (48). Machine learning (ML) and deep learning (DL) have recently improved predictions by analyzing large pharmacological datasets and uncovering complex patterns (49–55). For instance, Tropsha *et al.* used K-nearest neighbors (KNN) and support vector machine (SVM) algorithms to enhance logBB predictions (56–58). Recently, several models have been developed to predict BBB computationally. For instance, Mazumdar *et al.*, (59) developed a deep neural networks (DNN) model combining balanced Molecular ACCess System (MACCS) fingerprint with machine learning, achieving high accuracy (97.8%) in predicting BBB permeability with a dataset of ~8000 compounds. Another group of researchers (60) employed lightGBM architectures for logBB prediction, demonstrating significant improvements over other architectures (RF, kNN, MLR, SVM, AdaBoost etc.), finally reaching an accuracy of 0.85 (61). Additionally, Storch *et al.* introduced a multi-modal approach that integrates novel cheminformatics descriptors—GRID, providing a comprehensive framework for BBB prediction. It achieved a reported AUC of ~0.9 for the external test set. Most of the current research only reports accuracy, which is not enough to evaluate the model's performance. Classification models, which categorize compounds as BBB-permeable (CNS+) or non-permeable (CNS-), provide a practical way to integrate data from multiple studies despite varying experimental standards (50–54, 62–75). Modern DL methods like RNNs and CNNs further enhance predictions, with RNNs analyzing molecular fingerprints as sequences and CNNs interpreting 2D molecular images (72).

Graph Neural Networks (GNNs) have emerged as a cutting-edge approach for modeling molecular structures (76–78). GNNs represent deep learning on graphs and are approaches for semi-supervised learning on graph-structured data. Unlike traditional machine learning models that require extensive feature engineering, GNNs can automatically learn from the raw graph representation of molecules; therefore, they are not limited to the selection of specific descriptors that were manually developed from past experience. As a specific subtype of GNNs, Graph Convolutional Networks (GCNs) have become a focal point of interest recently, primarily due to their exceptional capability in processing data organized in graph structures (77). It uses convolutional operations to propagate information between nodes in a graph and is based on an efficient variant of convolutional neural networks that operate directly on graphs (79–81). This ability gives them a distinct advantage over traditional machine learning models in extracting more information on the topological aspect of data. The structure and interactions

of small molecules can be naturally represented as graphs, where atoms are nodes and bonds are edges. New architectures have been developed to enhance molecular property prediction (78). This representation fits perfectly with the architecture of GNNs and GCNs, leading to their increasing adoption in computational drug discovery. These models can capture the intricate relationship between molecular structure and function, leading to more accurate predictions and novel insights. As technology matures and integrates further with experimental methods, drug discovery is expected to become faster, cheaper, and more effective. Therefore, by representing molecules as graphs with atoms as nodes and bonds as edges, GNNs can capture intricate structural relationships that influence BBB permeability (82).

In this study, we introduced a novel *in silico* framework that utilizes GNNs to assess the BBB permeability of small drug molecules. We also investigated various machine learning/deep learning-based methods currently employed for BBB prediction for comparative analysis. The results demonstrated that the GNNs-based model of BBB pharmacanalytics exhibited superior performance compared to other models.

Methods

Data Source and Preparation

Chemical information with known BBB permeability properties of small molecule compounds/drugs datasets were collected from publicly available databases, including ChEMBL and PubChem (68, 83–85), which comprised a total of 1924 cleaned small molecular compounds and their assay data, including 1465 high penetration rate (BBB+) compounds and 353 low penetration rate (BBB-) compounds. BBB permeability is modeled as a classification task in which a molecule will be categorized into a BBB+ class or a BBB- class (Supplemental Tables I and II). The classification of BBB+ and BBB- was based on the BB ratio, which represents the ratio of total steady-state concentration in the brain to that in peripheral blood. Compounds assigned BBB+ should have a sufficiently high penetration rate (with a BB ratio ≥ 0.1) (86) or have been reported as CNS drugs or drug candidates under clinical development. In contrast, compounds assigned as BBB- should have a relatively low penetration rate (with a BB ratio < 0.1) or have been reported not to cross the BBB. The chemical structures were retrieved from the SMILES string and then converted to the SDF format using the RDKit toolkit (87). The 3D conformations of the compounds were generated and optimized using the LigPrep package of the Schrodinger software (88), with the protonation or deprotonation states of compounds calculated in pH 7.4. The dataset was split into the training and test sets (training: test = 3:1)

using a diverse selection to reduce the biases from the training set selection. The entire dataset was clustered based on the chemical diversity represented by molecular fingerprints. The compounds in each cluster were selected as the training compounds until reaching the required selection number for the training set, and the remaining compounds were collected into the test set. In this way, the selected training set was expected to uniformly cover the chemical space distribution of the entire dataset, which may be considered an ideal representative subset of the original dataset.

Overall Workflow

The schematic illustration of the workflow of this study is shown in Fig. 2. Compounds with the experimental label of BBB+ or BBB- were extracted and cleaned from multiple public resources. Four feature sets were calculated for

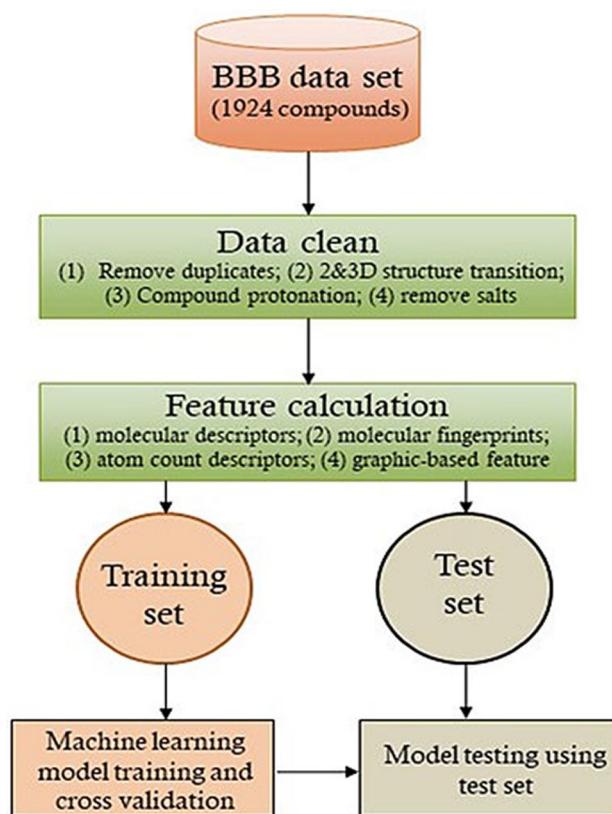


Fig. 2 The overall workflow for data processing and model training for BBB permeability prediction. A total of 1924 compounds were used in the study. After cleaning and pre-processing, different feature sets were calculated; the molecular descriptor, fingerprint, and atom count descriptors were used for traditional machine learning, and the graph-based features were used in GNNs models. The training and test sets were separated randomly with a ratio of 3:1. The machine learning models were trained using either one set of features or all 3 to evaluate the performance. The GNNs only use graph-based features

the entire compound sets, including molecular descriptors, molecular fingerprints, atom count descriptors, and atom-type-based graphic features. Six supervised machine-learning algorithms were applied to build classifiers for each prepared training set. Different types of features can describe and evaluate the properties of compounds from diverse aspects, and other machine learning algorithms may favor distinctive data structures. All the models were evaluated and compared to obtain the best model.

Molecular Descriptor Calculation and Selection

Four molecular descriptors were adopted to represent the molecular structure of all compounds, including physicochemical molecular descriptors, molecular fingerprints, atom type count descriptors, and graph-based molecular representation.

- *Physicochemical molecular descriptors.* 119 molecular descriptors were calculated using the RDKit toolkit (87), including MW, SlogP, TPSA, NumHBD, NumHBA, and others. For molecular fingerprint calculation, five types of different molecular fingerprints were collected using cheminformatics tools (85). Those descriptors were used in the traditional ML-based models for comparison.
- *MACCS and PubChem molecular fingerprints.* Molecular ACCess System (MACCS) fingerprint (89, 90) contains 166 binary fingerprints as substructure keys, each indicating the presence of one of the 166 MACCS substructure keys calculated from the molecular graph. The Extended Connectivity Fingerprints (ECFP) is a circular topological fingerprint with 1024 descriptors, representing the presence of particular substructures through circular atom neighborhoods (89). The Functional-Class Fingerprints (FCFP) is a variation of ECFP, which is further abstracted, instead of indexing a specific atom in the environment (89). The PubChem fingerprint covers a collection of 881 diverse substructure keys designed and used by PubChem (91, 92). The atom pairs fingerprint contains 1024 bits of binary data collected based on the atomic environments and shortest path separations of every atom pair in the molecule (93). Those descriptors were used in the traditional ML-based models for comparison.
- *Atom-type count descriptors.* The amber force field software toolkit calculated the atom type count descriptor, which defines the specific atom types and counts the total number of atoms belonging to each atom type (94). All the descriptors with low variance (<0.05) were removed during data pre-processing and cleaning. To avoid over-fitting issues, a feature-eliminating process was performed using L1 regularization (Lasso) to eliminate less correlated descriptors (95).

Of note, all of the above molecular descriptors were used separately or collectively to construct classifiers using traditional machine learning.

- *The graph-based molecular representation.* The graph-based molecular representation was created specifically for generating graph neural network models (Fig. 2). The graph-based molecular representation of each node in the molecular graph (atom types) was learned using embedding technology. Similar to the embedding methods in the natural language process (96), the input of the embedding model was the collection of one-hot-vectors for each atom type surrounding a particular atom type in a molecule, and the output was the one-hot-vectors of the center atom. The weights between the input and the first hidden layers were extracted from the initiated embedding vector for each atom type. Those embedding vectors were optimized by maximizing the conditional probability of predicting the central atom type correctly, given the information of surrounding atom types. Here, deep neural network (DNN) embedding models were built to learn the mathematical representation of a total of 53 predefined atom types that were initially defined in the general AMBER force field (GAFF) and designed for rational small molecular drug discovery (94). The training data set contained 123,182 drug-like compounds initially collected from the ZINC database (97). A different number of hidden layers (1, 2, 3, 4, 5, 6) in the DNN model were tested, and 4 hidden layers were selected, achieving an accuracy of 93% in the tenfold cross-validation. The entire atom embedding section was processed using Python code (Python 3.7.6), and the DNN models were developed using the *TensorFlow* package (Tensorflow 2.1.0) (98).

Model Development With Machine Learning and Deep Learning Frameworks

A prediction pipeline was developed for supervised classification with various machine learning algorithms, including Random Forest (RF) (99), SVM (100), AdaBoost decision tree (AdaBoost) (101), Naïve Bayes (NB) (102), Logistic Regression (LR) (103), and Neural Network/MultiLayer Perceptron (NN/MLP) (104). The open-source Python module Scikit-learn was used for model training, data prediction, and result interpretation (105). For RF, the RandomForestClassifier function from Scikit-learn was applied. The model was saved after optimizing parameters *n_estimators* (50, 100, 500) and *max_depth* (2, 3, 4, 5). The method was applied to SVM with three kernel functions (linear, rbf, poly). The SVM model with the best performance was saved after the optimization on penalty parameter *C* and parameter γ for rbf and poly kernels. For AdaBoost, the AdaBoostClassifier

function was applied to optimize parameters `n_estimators` (10, 100, 1000) and `learning_rate` (0.01, 0.1, 1). The weaker classifier used in AdaBoost was set to `DecisionTreeClassifier`. When training NB models, the `BernoulliNB` method was applied to datasets with fingerprints as features. Given that `Bernoulli Naïve Bayes` requires binary-valued feature vectors for samples, the prior probabilities of the classes were set to none. For MLP, the `MLPClassifier` method was applied with the setting of different hidden layers (1–5). The following parameters were also optimized during the model training: activation function (identity, logistic, tanh, relu) and learning rate (0.1, 0.01, 0.001, 0.0001). The `LogisticRegression` was applied to implement the logistic regression model with L1 penalty (lasso). The parameter solver was set to sag to handle the multinomial loss in large datasets. The hyper-parameters tuning of each estimator was performed using the `GridSearchCV` tools, and the best-performance models in the cross-validation were selected. To make a balance between 2 positive and negative predictions, the weights associated with classes (BBB+/BBB-) were used to minimize the in-balance between 2 classes in the dataset.

Moreover, graph-based deep neural network models were constructed and compared in this study. The graph-based molecular feature was applied as the input data using the embedded vector for each atom type. The topology and connectivity of all the atom types for a molecule were extracted using the adjacency matrix. Each atom in the molecule and its neighboring atoms can be identified using the adjacency matrix. The atomic features of the corresponding bit and

its neighboring atoms will be added up and projected to another space. Collectively, four types of GNNs architectures were adopted to construct the models: 1) convolutional graph neural networks using the adjacency matrix as the filter (GCN_1); 2) convolutional graph neural networks using normalized Laplacian matrix as the filter (GCN_2); 3) convolutional graph neural networks using Chebyshev polynomials of the Laplacian matrix as the kernel (GCN_3); and 4) graph attention neural networks (GAT) (Fig. 3). In addition, the drop-out and pooling layers were added after the GNNs layers to optimize the model and avoid overfitting. After the GNNs and pool layers, the resultant feature vectors of each atom will be flattened into a 1D vector using fully connected (FC) hidden layers. Then, a Softmax function component will be added to the architecture for classification.

Model Evaluation

Ten-fold cross-validation was performed for model generation and evaluation. The model was trained using any nine folds as training data, and the resulting model is validated on the remaining fold of data. Different statistical metrics were calculated to evaluate the performance of machine learning models from diverse aspects. The area under the receiver operating characteristic curve (AUROC) was calculated after the acquired true- and false-positive rates. AUROC can be indicated as assessing the model model's performance on separating classes. At the same time, the Balanced F-score was calculated as the weighted average of precision and recall.

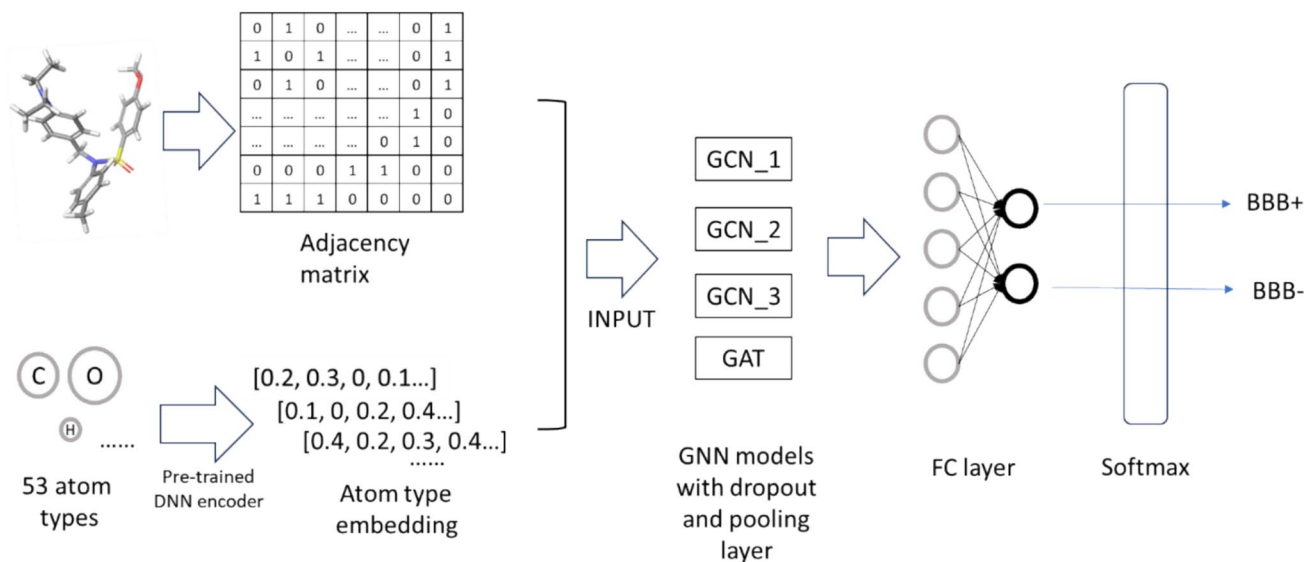


Fig. 3 A schematic diagram of the study design of building GNNs models. Each compound will be represented by an adjacency matrix based on the connectivity of all atoms, and each atom type will be represented by an embedding vector through a pre-trained deep-learning neural network encoder. Those data will be input into the graph

convolutional networks (GCN) or graph attention network (GAT), GCN_1 to GCN_3, using different filters/kernels. A full connect (FC) layer and a SoftMax function are placed after the GNNs models for classification

Precision measures the accuracy of positive predictions, while recall measures the ability to identify all relevant instances. The accuracy classification score (ACC) was calculated to compute subset accuracy, verifying whether the label predicted for one sample matches the corresponding actual value. In addition, Matthew's correlation coefficient (MCC) was calculated to measure the quality of binary and multi-class classifications. The MCC score is a balanced measure in which both the true and false positives and negatives are considered. The detailed equations are shown in the table below (Table I):

Results

Prediction Evaluation in Cross-Validation

The ROC AUC value of all the machine learning models for the cross-validation is summarized in Table II. Models performed relatively consistently when using the same types of molecular descriptors. Overall, the four GNNs models (GCN_1, GCN_2, GCN3, and GAT) outperformed all the other algorithms, with the highest predictive accuracy reaching 0.95. RF and MLP performed better than the other methods of traditional machine learning. From the perspective of molecular descriptors, besides graph-based molecular

representation used for GNNs, molecular fingerprints, especially FCFP, provided more information that contributed to the improved ML model of the ML programs. Notably, the RF model combined with FCFP descriptors as features achieved comparable performance to the GNNs models.

Figure 4 shows the ROC plots for the four best models from cross-validation. Two use GNNs methods (GCN_3 and GAT) and traditional machine learning methods (RF and MLP). Those results indicate that all the best-performance models made a reliable prediction with promising sensitivity and specificity values.

Model Evaluation with Metrics

Instead of using only the ROC AUC score, a series of metrics were calculated to explore further the performance of each machine learning algorithm on different feature types. Metrics functions assess prediction errors for specific purposes and can evaluate the model performance from various aspects. The other metrics involved in this study are F1 score, Accuracy, MCC, precision, and recall. For each algorithm, the best-performed model was evaluated using the test set.

As the results in Fig. 5 show, the test dataset performance was relatively consistent with the cross-validation result in Fig. 2. GCN_2 model made the most accurate prediction on the test dataset evaluation. At the same time, the RF and MLP models also performed well, with most of the statistical evaluation metrics aligning with the ROC AUC when comparing different algorithms (Fig. 4). Noticeably, the linear regression model (LR) combined with lasso regularization achieved a relatively good predictive performance over the test set. However, the GAT model's performance on the test sets was comparable to its performance during training. Overall, the GNNs models perform better than traditional machine learning methods. It is worth mentioning that the recall scores of all GNNs models were significantly better than ML models. Traditional ML methods like RF have relatively high

Table I Equations for Classification Model Evaluation Metrics

Metric	Equation
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$
F1 score	$(2 * TP) / (2 * TP + FP + FN)$
Matthews correlation coefficient (MCC)	$(TP * TN - FP * FN) / \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$

*FN, number of false negatives; FP, number of false positives; TN, number of true negatives; TP, number of true positives

Table II AUROC of All Machine Learning Models on Cross-Validation

Descriptor	LR	SVM	NB	RF	AB	MLP	GCN_1	GCN_2	GCN3	GAT
Atom count	0.9006	0.9069	0.8344	0.9386	0.916	0.9286				
RDKit	0.915	0.8995	0.8636	0.9297	0.9316	0.9042				
MACCS	0.8953	0.8577	0.8363	0.9237	0.9034	0.837				
PubChem	0.9135	0.8743	0.8975	0.926	0.926	0.902				
ECFP6	0.9193	0.939	0.8222	0.9247	0.9005	0.9283				
FCFP6	0.9313	0.8902	0.843	0.9489	0.9384	0.9239				
All (Lasso)	0.9371	0.8879	0.8512	0.9332	0.9387	0.9422				
Mol Graph							0.936	0.9423	0.9424	0.9504

*LR, logistic regression; SVM, support vector machine; RF, random forest; AB, adaptive boosting; MLP, multi-layer perceptron; GCN, graph convolutional network; GAT, graph attention network; MACCS, Molecular ACCess System; ECFP, Extended Connectivity fingerprints; FCFP, Functional Class fingerprints

Fig. 4 The ROC plot of the selected best models from cross-validation. This figure shows the best 4 models in the training phase with an AUCROC value of tenfold cross-validation; the values in the figure represent the average values of 10 runs. (TPR: true positive rate, FPR: false positive rate)

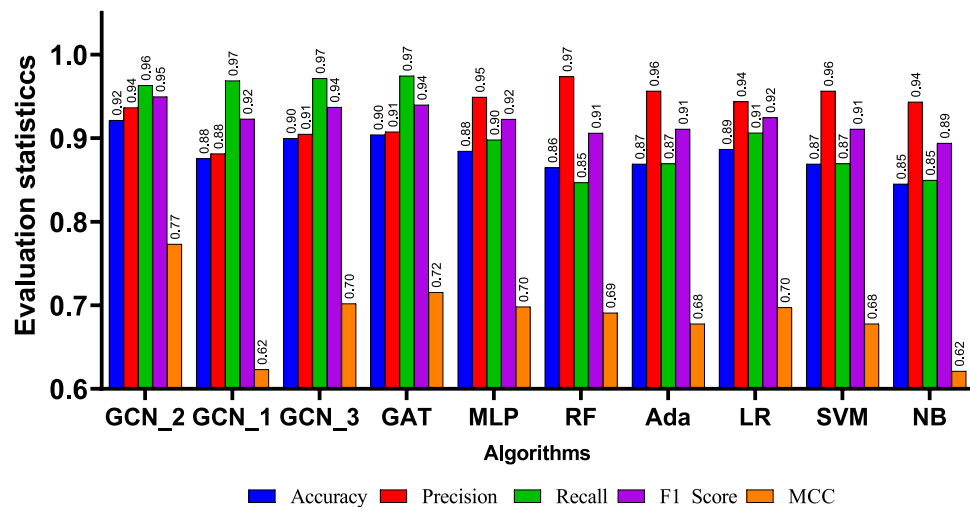
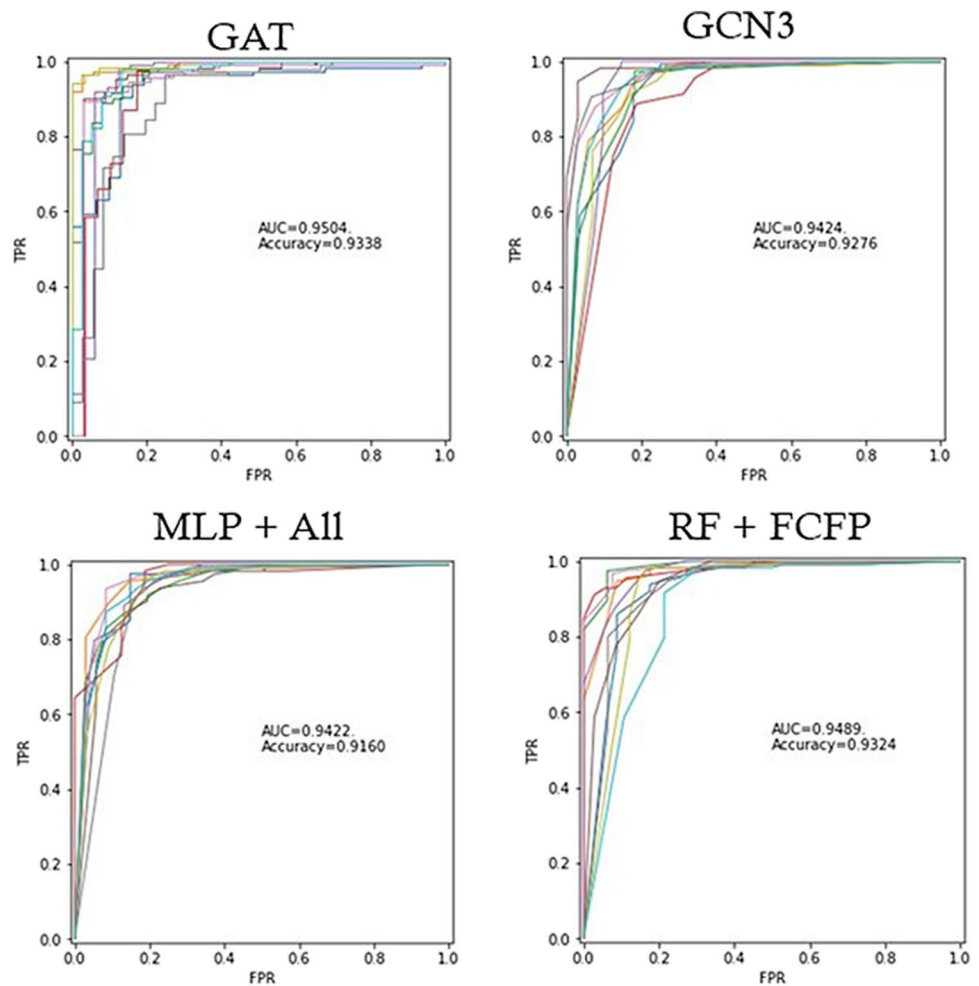


Fig. 5 Predictive performance metrics on the test set over all the models. Accuracy, precision, recall, F1 score, and MCC score were compared across all models. The GCN_2 (convolutional graph neural networks using normalized Laplacian matrix as the filter) was the best among the 4 GNNs models and superior to other ML algorithms.

When tested against the test dataset, it achieved an accuracy of 0.92, precision of 0.94, recall of 0.96, F1 score of 0.95, and MCC score of 0.77. Note, the overall recall scores of the GNNs models were better than those of traditional ML methods

precision scores, but the imbalance between precision and recall makes it less valuable than the GNNs model.

This difference holds considerable significance within the clinical context, especially in the realm of drug discovery and development. Specifically, in the evaluation of a drugs' ability to cross the BBB, accurately classifying a substance as BBB+ or BBB- is crucial. Here, the risk associated with falsely identifying a BBB+ drug as BBB- (a false negative) is considerably more critical than the converse (106). Such errors can lead to the overlooking of potentially effective therapeutic agents for neurological conditions, thereby hindering advancements in treatment options.

Discussion

BBB permeability prediction is critical for drug discovery for both CNS and non-CNS targeting drugs; for non-CNS drugs, predicting BBB permeability can help identify compounds that might inadvertently cross the BBB and cause unwanted CNS side effects. For CNS-targeted drugs, particularly those addressing neurological disorders like Alzheimer's disease (AD), ensuring effective BBB penetration is essential for therapeutic efficacy. Beyond the scope of AD drug discovery, the ability to accurately predict BBB permeability is equally crucial for developing treatments targeting a broad spectrum of central nervous system (CNS) conditions. Although *in vitro* and *in vivo* assays have long been used to measure BBB permeability, they are very time- and effort-consuming. Computational pharmacanalytics models can rapidly screen large compound libraries, helping to identify potential lead molecules.

In this research paper, we developed a framework of GNNs for predicting BBB permeability with graph representation of small molecules with atom-type embedding. Our results show that while traditional machine learning models can already achieve a relatively good model performance in BBB prediction, GCNs bring a set of unique capabilities tailored for graph data and achieve even better performance. Their unique architectures are dedicated to capturing complex patterns in graph-structured small molecule datasets, outperforming traditional models in the BBB permeability prediction task. Out of four different frameworks, in terms of the overall performance, including F1 score, Accuracy, MCC, precision, and recall, we found convolutional graph neural networks using Chebyshev polynomials of the Laplacian matrix as the kernel (GCN_2) have the best performance (107). Thus, we have named it "GCN_BBB". The reason why we think this model performed best across all GNNs models is because by the Chebyshev transformation of the Laplacian matrix, more information of the connectivity between atoms were extracted, giving the GNNs model more information to build the inference between the

molecular structure and the activity. Generally, our GCN_BBB model can be especially useful in drug lead development and QSAR study design as it can screen many compounds relatively quickly and provide information on how to design the molecule more effectively.

There are a few advantages and innovations to using the GCN model for BBB prediction, such as the leveraging of relational information inherent in molecular structures by treating atoms as nodes and bonds as edges in a graph. This allows for the direct incorporation of 3D molecular structure into the predictive model, providing a more nuanced understanding of how different chemical groups affect BBB permeability (108). Our GCN_BBB model differentiates itself by leveraging the unique capabilities of GNNs, particularly GCNs using Chebyshev polynomials of the Laplacian matrix, to specifically address the challenges of graph-structured data in drug discovery. Therefore, GCN_BBB emphasizes the automatic learning of molecular features directly from graph representations without the need for extensive feature engineering or additional data modalities, streamlining the predictive process and potentially uncovering new insights into BBB permeability. This positions our model as a complementary tool alongside these cutting-edge approaches, enhancing the overall predictive landscape for BBB permeability in drug discovery.

Another interesting finding is that the GAT model shows the best performance in the training phase while the GCN_2 outperformed GAT in the test data set. This might indicate that GAT over-fits due to its large number of parameters; as the network gets deeper (more layers) or broader (more neurons per layer) or if we increase the number of attention heads, the number of parameters grows, which can increase the model's representational power at the cost of requiring more data to effectively learn without overfitting. Comparing to published methods discussed in the introduction section, our GCN_BBB model differentiates itself by leveraging the unique capabilities of GNNs, particularly GCNs using Chebyshev polynomials of the Laplacian matrix, to specifically address the challenges of graph-structured data in drug discovery.

In the realm of drug discovery and development, ensuring a high recall score is of paramount importance (109). A high recall score indicates a model's ability to minimize the incidence of false negatives. In practical terms, this means that GNNs models are more adept at correctly identifying drugs that are capable of crossing the BBB, thereby reducing the likelihood of erroneously excluding potential CNS therapeutic agents. This capability is particularly valuable in the early stages of drug development, where the cost of missing a promising compound can be substantial, not only in terms of financial resources but also in delayed treatment options for patients.

Moreover, the emphasis on recall scores reflects a cautious approach in medical research, prioritizing safety and the minimization of oversight in identifying viable drug candidates (110). GNNs models, with their superior recall scores, thus represent a significant advancement in the field, offering a more reliable toolset for navigating the complexities of drug development. Their application promises to enhance the efficiency and accuracy of the drug discovery process, highlighting their critical role in bringing safer, more effective treatments during early drug development with fewer risks of overlooking potential breakthroughs.

Nevertheless, this study has several limitations: 1) The developed model relies on existing data and might not be accurate for novel chemical spaces. We trained our model using a meticulously curated and verified dataset, ensuring only high-quality data with direct citations from reliable experimental studies were included. In the future, we plan to incorporate additional data to enhance model performance by increasing diversity and minimizing overfitting. To enhance scalability and generalizability, a hybrid approach that combines computational predictions with select experimental validation is optimal for ensuring both efficiency and accuracy in assessing BBB permeability. 2) Our model focuses on BBB permeability prediction of small molecules (<500 Da); it may not be suitable for predicting macromolecules passing BBB through a transporter or receptor-mediated transcytosis pathway. 3) The BBB is not just a physical barrier; it's also a metabolic one, with enzymes that can modify or degrade drugs. The BBB function in some neurological diseases is altered; for example, in AD, the function of the efflux pump can be severely dysfunctional, which partly accounts for A β accumulation in the CNS system (111). Thus, our model only considers the general conditions but not the specific scenarios of certain diseases. In addition, in terms of the model interpretation, our GNNs models mostly perform like a black box and have a limitation of interpretation, which is the same with most deep learning models. And there are still lack of capacity from the computer science and mathematics side to provide more insights into how the model works and makes good predictions.

As for future direction, we aim to develop disease-specific and personalized BBB permeability models. We recognize that chronic health conditions, notably hypertension (112) and diabetes (113), significantly impact BBB integrity and function. Such models promise to advance our understanding of disease pathophysiology and offer the potential for targeted therapeutic strategies, ensuring a more individualized approach to patient care and drug delivery to the CNS system. In another direction, we would like to develop a more interpretable model, providing recommendations on lead compound optimization – i.e., how to modify a given molecule to increase or decrease the BBB permeability potential.

Conclusion

This study introduces an ML/DL classification framework for predicting BBB drug permeability. The efficacy of the deep graph neural network methods in predicting the drug's BBB penetration was compared with the existing methods. The prediction accuracy of the best mode convolutional graph neural networks using normalized Laplacian matrix as the filter (GCN_2) over the external test datasets already achieved 0.92, precision of 0.94, recall of 0.96, F1 score of 0.95 and MCC score of 0.77. Furthermore, our machine learning methods' accuracy, AUC, and F1 scores with both the training and test sets are relatively consistent, indicating that our model is stable and not overfitting. We are among the first researchers to use graph neural networks for small compound BBB permeability prediction. The result indicates that the graph-based deep learning model has improved prediction performance over traditional machine learning methods. In the future, we will further evaluate our model using more experimental data and endeavor to apply our method and technology to facilitate CNS drug discovery.

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Data Availability Data files used in this work can be found in Supplementary Information. Additional information is available upon request.

Declarations

Conflict of Interest All authors have no conflict of interest to declare.

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